Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Mathematics I**

Code No: Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Solve the inequality $6 + 5x - x^2 \geq 0$.

3. Find the domain and range of the function $f(x) = \sqrt{6 - x - x^2}$.

4. Prove that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 3x - 1$ is bijective.

5. Expand e^x about $x = 0$ by using the Maclaurin series.

6. Find the inverse matrix of the matrix $\begin{pmatrix} 1 & 4 & 1 \\ 3 & 3 & -2 \\ 0 & -4 & 1 \end{pmatrix}$.

7. There are 7 men and 3 ladies. Find the number of ways in which a committee of 6 persons can be formed if the committee should have at least one lady.

8. Find the equation of a parabola having vertex (+, 2) and directrix $x = 4$.

Group C

Attempt any TWO questions[2x10=20]

9. a) If A and B be two subsets of universal set U such that $n(U) = 350$, $n(A) = 100$, $n(B) = 150$, and $n(A \cap B) = 50$, then find $n(\overline{A} \cap \overline{B})$.b) If a, b, c are in A.P., b, c, d are in G.P., and c, d, e are in H.P., then prove that a, c, e are in G.P.

10. a) Prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a - b)(b - c)(c - a)$.b) By using the vector method, prove that $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

11. a) Define a parabola with different parts using the figure and derive the standard equation of parabola $y^2 = 4ax$.b) In how many ways can the letters of the word “ARRANGE” be arranged so that all the vowels are always together?